Paleomagnetism and age of the Aston dykes and Savage Point sills of the Boothia Uplift, Canada

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Oriented samples (107) have been drilled for paleomagnetic studies from diabase sills and dykes on either side of the Boothia Uplift, Canada. Six sites at Savage Point are in sills which intrude the Aston Formation; three, at Whitehead Point, north of Savage Point on the east coast of Prince of Wales Island, are in sills which also intrude these strata. Six sites in the Aston Bay area of Somerset Island are in dykes and a sill which cut the Aston Formation; the former also intrude the Hunting Formation. No meaningful results were obtained from the Whitehead Point samples. Paleomagnetic measurements on the Savage Point samples, together with new K–Ar data, strongly suggest these sills belong to the Mackenzie igneous episode (ca. 1240 Ma). The sill in the Aston Bay area apparently belongs to this episode too, whereas most of the dykes are Franklin dykes (ca. 675 Ma). One large dyke was possibly emplaced in Tertiary times. Thus the Aston Formation is older than 1240 Ma and the Hunting Formation is bracketed between 1240 and 675 Ma.

On a obtenu par forage 107 échantillons orientés pour études paléomagnétiques dans les sills et les dykes de diabase de part et d'autre du soulèvement de Boothia, Canada. Six localités à Savage Point sont dans des sills qui recoupent la formation d'Aston; trois localités, à Whitehead Point, au nord de Savage Point sur la côte est de l'île de Prince de Galles, sont dans des sills qui aussi recoupent ces strates. Six localités dans la région d'Aston Bay sur l'île de Somerset sont dans des dykes et un sill qui coupent la formation d'Aston; les dykes pénètrent aussi la formation de Hunting. On n'a obtenu aucun résultat significatif des échantillons de Whitehead Point. Les mesures paléomagnétiques sur les échantillons de Savage Point en plus des données K–Ar nouvelles, suggèrent fortement que les sills appartiennent à l'épisode intrusif de Mackenzie (environ 1240 Ma). Le sill dans la région d'Aston Bay appartient apparemment à cet épisode aussi, alors que la plupart des dykes sont des dykes de Franklin (environ 675 Ma). Un gros dyke a probablement été mis en place au Tertiaire. Ainsi, la formation d'Aston serait plus ancienne que 1240 Ma et la formation de Hunting daterait de la période entre 1240 et 675 Ma.
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Introduction

Paleomagnetic studies of basic intrusive rocks of the Canadian Shield have contributed a great deal in recent years to the correlation of these units and to the determination of their ages. The paleomagnetism of the intrusives has also helped in correlating and dating supracrustal sequences by providing minimum, and sometimes maximum, ages for these sequences.

Diabase dykes and sills commonly intrude the Upper Proterozoic strata that occur on the islands and peninsulas just north of the main part of the Canadian Shield — for example, on Baffin, Somerset, Prince of Wales, and Victoria Islands. More than one series of intrusions are present in this region so their paleomagnetism could be an important parameter in determining their correlation and the correlation and tectonic history of the supracrustals which they intrude. Paleomagnetic results are already available from Franklin dykes (K–Ar age 675 Ma) which cut the Uluksan and Eqalulik Groups on Baffin Island (Fahrig *et al.* 1971; Fahrig and Schwarz 1973; Jackson *et al.* 1975) and from diabase sills and dykes of similar age that intrude the Shaler Group and Natkusiak Formation in the western part of Victoria Island (Thorsteinsson and Tozer 1962; Palmer and Hayatsu 1975). The present study is designed to extend this work to the dykes and sills that cut Upper Proterozoic strata in the Boothia region.

Sampling

Sills on Savage and Whitehead Points on the east coast of Prince of Wales Island and a number of

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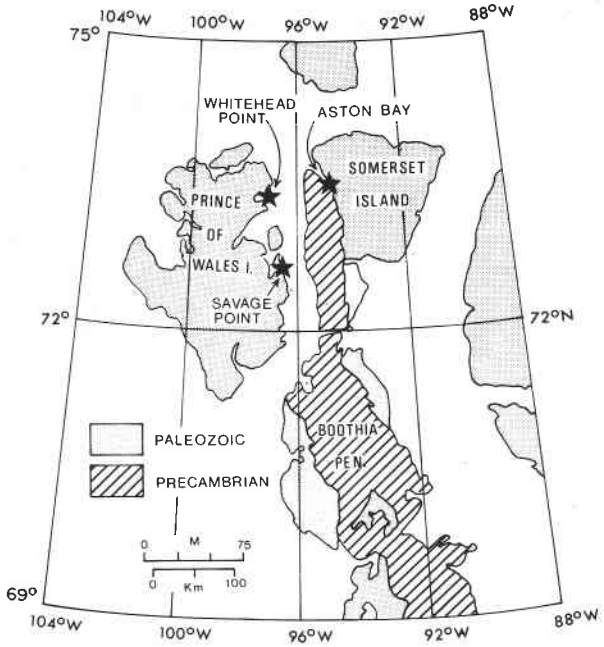


FIG. 1. Index map showing the three sampling areas on the Boothia Uplift.

dykes and one sill in the Aston Bay area of Somerset Island were sampled (Fig. 1). From six to eight drilled cores (2.5 cm in diameter) were obtained at each site and each core (sample) was oriented by means of a sun compass. Contacts between diabase and intruded sediments were not located and thus material for contact tests could not be obtained.

The distribution of diabase and of sampling sites in the Savage Point area is shown in Fig. 2*a*. The number of individual sills present in this area is not known and it is possible that all of the basic rock here represents a single thick sill. Site 2, for example, is ultrabasic material, probably from an olivine cumulate layer near the base of the sill. In contrast, site 4 is extremely coarse-grained, granophyric, hornblendic material which may represent the upper volatile-rich phase of the sill. Sites 10 and 11 are from the lower and central parts of a thick sill. Whole-rock K–Ar ages have provided a first approximation to the ages of the Savage Point sills. These are 830 ± 29 Ma for material from site 7, and 1084 ± 83 Ma for material from site 10 (R. K. Wanless, personal communication, 1975). These should be treated as minimum ages for the intrusive event(s) and suggest that the diabases are not part of the Franklin intrusions. Kerr and Christie (1965) have demonstrated that most of the tectonic movement along the Boothia Uplift was Phanerozoic. Paleomagnetic directions for the diabase must therefore be corrected for tilt; the amount is indicated by the tilt of the Upper Proterozoic Aston Formation which dips to the west at 5–25° in the Savage Point area.

The location of samples obtained at three sites on a diabase sill (or sills) on Whitehead Point is shown on Fig. 2*b*. The Aston Formation is almost flat-lying at this locality.

Five dykes and a sheet-like mass of diabase were sampled in the Hunting River (Aston Bay) area of Somerset Island (Fig. 2*c*). The dykes all cut the Hunting Formation which is the uppermost Proterozoic unit on Somerset Island (Fortier *et al.* 1963; Kerr and Christie 1965; Dixon 1974). This formation dips to the northeast at 10–35° in the sampled area. The dyke material is fresh and, except for site 27, is feldsparphyric. Site 27 is from a dyke with an exposed width of several hundred feet; its contacts were not observed. It consists almost entirely of red granophyric material which probably indicates that the present erosion surface is not a great deal below the original upper limit of the dyke. Site 21 is from the sill-like mass of rust weathering diabase (the dykes generally weather dark olive). The diabase exhibits hematite alteration and hematite stain along cracks. It is not clear whether it intrudes the Hunting as well as the underlying Aston Formation. Kerr (1977) has indicated that the Hunting overlies the Aston unconformably and diabase both pre-dates and post-dates the unconformity. Site 21 may be on diabase truncated by the unconformity, hence the hematization. Site 21 is in the general area from which material was obtained by Dixon for a K–Ar age of 702 ± 25 Ma (Dixon 1974). However, the material collected by Dixon was much fresher and may be from a younger nearby dyke.

Paleomagnetic Methods and Results

General Method

Specimens of length 2.5 cm were cut from the bottom of the cores and measurements of their natural remanent magnetization (nrm) were made using predominantly a biastatic magnetometer but occasionally, when specimens were weakly magnetized (for example, those from site 27), a JR-3 spinner magnetometer. At least two specimens (each from a different sample) per site were demagnetized progressively in alternating magnetic fields generally in steps of 5 mT (peak) to 40 mT (peak) using either two-axis tumbling equipment or a Schonstedt single-axis demagnetizer. Whenever both specimens gave good end-points the remaining specimens from that site were demagnetized in several fields within the end-point region. The site mean direction associated with the highest precision is regarded as the best estimate of the cleaned direction of magnetization.

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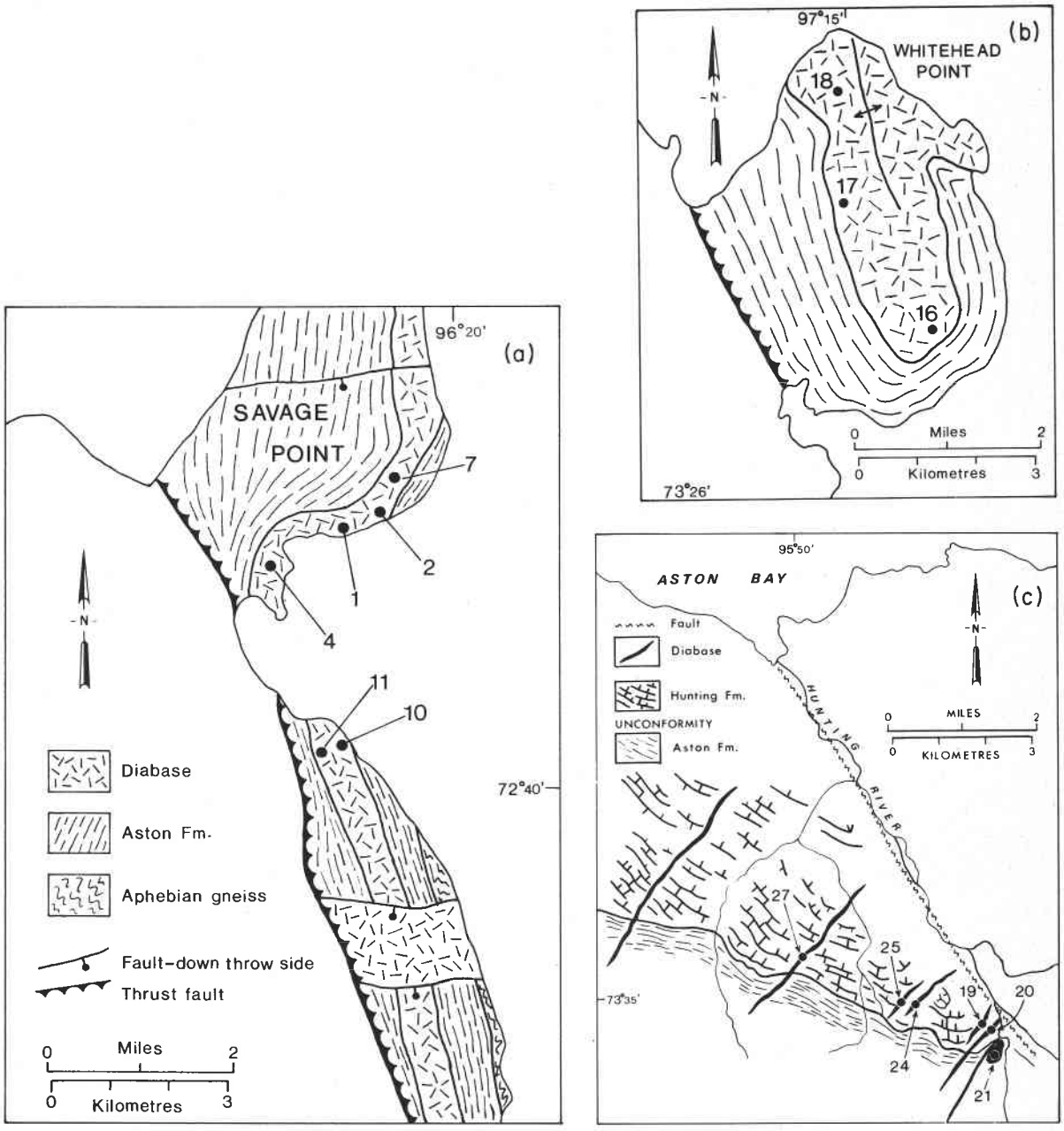


FIG. 2. Detailed geology of the three sampling areas. Geology of Savage and Whitehead Points (*a*, *b*) modified slightly from Christie (1971). Geology of Aston Bay area (*c*) modified slightly from Dixon (1974).

Savage Point

Natural remanent magnetization sample directions within each site were generally well grouped, while site mean directions tend to be strung between the present earth's field direction (steeply down) and a roughly horizontal westerly direction (Table 1). Good end-points were obtained from all pilot specimens and, after cleaning the remaining specimens, it was only necessary to reject one sample from the analysis. This sample, from site 01, gave a stable direction more than 90° from those of the remaining samples from the site. This sample possibly was misoriented or drilled from a boulder which had rotated. The site mean directions after cleaning have all moved away from that of the earth's present field (Fig. 3) to a westerly, generally shallow dipping direction. The mean direction for the six sites is $D = 260.3°$, $I = 14.4°$ with $k = 21.5$

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TABLE 1. Site mean directions of magnetization for the Savage Point, Whitehead Point, and Aston Bay diabases

Region	Site	N	n	Natural remanent magnetization D	I	R	k	Alternating magnetic field demagnetization B_{af}	D	I	R	k	$\alpha_{95}°$
Savage Point	01	7	6	258	52	5.961	128	20	259.4	11.9	5.995	960	2.2
(Prince of Wales Is.)	02	7	7	251	34	6.509	12	15	256.6	9.3	6.878	49.4	8.7
	04	7	7	263	37	6.948	115	25	263.2	5.1	6.947	113	5.7
	07	8	8	254	47	7.305	10	15	255.5	17.6	7.995	1440	1.4
	10	7	7	263	0	6.991	695	10	265.5	−3.0	6.989	556	2.6
	11	8	8	269	75	7.992	90	20	262.2	46.9	7.902	71.3	6.6
Whitehead Point	16	7	7	61	82	6.881	50						
(Prince of Wales Is.)	17	7	6	354	81	5.847	33						
	18	7	7	30	84	6.994	940						
Aston Bay	19	7	7	95	66	6.833	36	20	88.0	24.3	6.955	135	5.2
(Somerset Is.)	20	7	7	98	22	6.923	78	15	99.5	10.2	6.920	75.3	7.0
	21	8	8	231	39	6.887	6	15*	237.3	−9.3	7.827	40.6	8.8
	24	6	6	106	58	5.938	81	15	111.2	41.9	5.922	64.2	8.4
	25	7	7	98	72	6.914	70						
	27	7	7	167	88	6.954	130	25	72.4	85.5	6.899	59.2	7.9

NOTES: N = number of samples (cores) collected; n = number of samples used in the analyses; D = declination; I = inclination; R = resultant of n unit vectors; k = Fisher's (1953) estimate of precision; $\alpha_{95}°$ = semi-angle of cone of 95% confidence about the mean direction; B_{af} = peak alternating magnetic field in mT. *The result from sample 01 in 55 mT has been used in computing the mean direction.

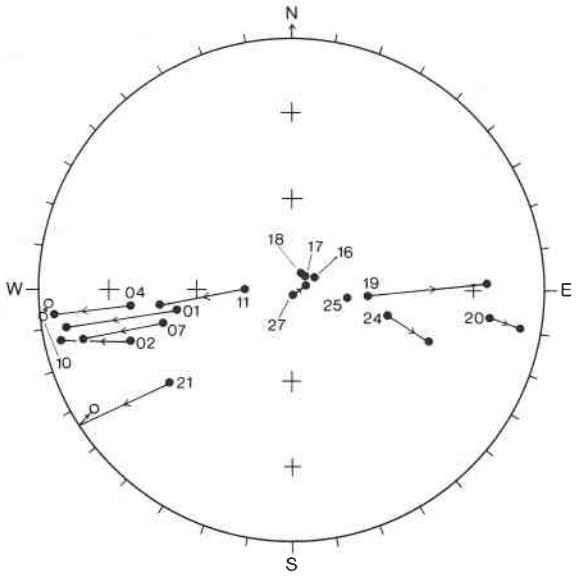


FIG. 3. Site mean directions before and after af demagnetization. Equal area projection. Solid circles are north-seeking directions down.

and $\alpha_{95} = 14.8°$ while the mean pole position is at 4.9°N and 185.1°E with $K = 53.5$ and $A_{95} = 9.2°$. The pole from the mean direction (PMD) is at 4.1°N and 184.9°E with $\delta p = 7.8°$ and $\delta m = 15.2°$. After correction for tilt (Table 2) the mean direction becomes $D = 260.4°$, $I = -0.3°$ ($k = 23.8$, $\alpha_{95} = 14.0°$) with a mean pole at 3.0°S and 182.6°E ($K = 75.0$, $A_{95} = 7.8°$) (PMD at 3.0°S, 182.7°E with $\delta p = 7.0°$ and $\delta m = 14.0°$). Although the increase in k (or K) is not significant at the 95% confidence level, it certainly does not contradict the contention that it is the mean direction after tilt correction that corresponds to the direction acquired by the diabase on intrusion.

Whitehead Point

Omitting the results from sample 3 of site 17, the nrm sample directions within each site were well grouped, with site mean directions very steeply down, presumably reflecting the influence of the earth's present field. The nrm intensity of sample 3 was very much greater than that of other samples from the site. Its direction was about 140° from the mean of the rest and was stably magnetized, and a lightning strike is suspected.

Alternating magnetic field (af) demagnetization of pilot specimens from site 16 was characterized by large erratic changes in direction and very rapid decreases in intensity; typically at 10 mT the magnetization had decreased to less than 5% of the nrm value. All samples were demagnetized in a variety of fields but useful results were not obtained. Except for sample 3, samples from site 17 generally maintained their nrm direction during demagnetization and the primary magnetization appears to have been completely obscured by a recent overprint. Similar results were obtained with five of the samples from site 18. The directions of magnetization of pilot specimens from samples 1 and 2 initially moved along great circles towards the east and northeast respectively, but end-points were not observed before the onset of instability. Thus, none of the diabase sites on Whitehead Point yielded meaningful results.

TABLE 2. Dip-corrected site mean directions and corresponding pole positions for the Savage Point and Aston Bay diabases

Region	Site	Beds		B_{af}	D	I	Pole		
		Strike	Dip				Lat.	Long.	Polarity
Savage Point	01	168	12W	20	259.4	0.1	3.1°S	176.3°W	N
(Prince of Wales Is.)	02	168	12W	15	256.6	−2.5	5.2°S	174.1°W	N
	04	168	12W	25	263.3	−6.7	5.2°S	179.0°E	N
	07	168	12W	15	255.6	5.8	1.5°S	171.9°W	N
	10	174	21W	10	265.7	−24.2	13.4°S	173.8°E	N
	11	174	21W	20	262.5	25.6	10.6°N	175.4°W	N
Aston Bay	19	020	15W	20	84.3	38.0	22.0°S	174.4°E	S
(Somerset Is.)	20	020	15W	15	98.6	24.9	10.1°S	163.3°E	S
	21	020	15W	15–55	234.3	−18.1	18.5°S	152.4°W	N
	24	084	10N	15	102.3	45.8	22.6°S	155.6°E	S
	27	130	25N	25	45.0	61.1	52.4°N	26.1°E	N

NOTE: Strike of the surrounding beds measured in degrees clockwise from due north; dip is positive below the horizontal; other symbols are defined in Table 1.

Aston Bay

Natural remanent magnetization sample directions from site 21, the diabase sill, were scattered between that of the earth's present field and a southwesterly, almost horizontally dipping, direction. Various magnetic characteristics were displayed on af demagnetization and it was necessary to treat each specimen individually. Figure 4*a* illustrates the directional behaviour of a number of specimens. Those from samples 1, 2, and 4 show that, here, three magnetic components are involved; the first to be removed is apparently due to the earth's present field. A second, magnetically harder component, is then gradually removed until the direction becomes stable at what is considered to be that of the primary magnetization. Sample 3 displays only the two harder components while the rest either display the primary magnetization *ab initio* or have an easily removed viscous component due to the earth's present field superimposed upon it. Sample directions after demagnetization form a close group (Table 1) with a mean to the southwest and gently dipping upwards. The harder of the two secondary magnetizations could be related to erosion prior to deposition of the Hunting Formation or to the later intrusion of a nearby dyke.

Good within-site grouping of nrm sample directions was observed for each of the five dyke sites (Fig. 2*c*) with site mean directions strung between a vertical down direction and a shallow-dipping *easterly* direction. Three of these sites responded well to af demagnetization (19, 20, and 24); in each case a component due to the earth's present field was removed (Table 1; Fig. 3). All specimens from site 25 were treated individually; initially their directions became less steeply dipping and easterly, suggesting that this dyke may be identified with those a little to the east. However, no end-points were observed before the onset of instability and no definitive result has been obtained. The mean direction for sites 19, 20, and 24 is $D = 98.8°$, $I = 25.7°$ with $k = 18.8$ and $\alpha_{95} = 29.3°$ giving a mean (S) pole at 11.0°S and 162.3°E with $K = 30.9$ and $A_{95} = 22.6°$ (PMD at 10.5°S, 163.0°E with $\delta p = 17.0°$ and $\delta m = 31.6°$). After correction for tilt of the surrounding beds (Table 2) the mean direction is $D = 95.0°$, $I = 36.5°$ with $k = 39.3$ and $\alpha_{95} = 19.9°$ with a mean pole at 18.4°S, 164.4°E ($K = 51.3$, $A_{95} = 17.4°$) (PMD at 18.0°S, 164.6°E with $\delta p = 13.5°$ and $\delta m = 23.2°$).

Site 27 is weakly magnetized (sample mean nrm intensity, 8.7×10^{-4} A/m (8.7×10^{-7} emu/cm^3)) almost vertically down. Directional movements of all specimens during progressive af demagnetization were generally small, and the site mean direction at each of the seven demagnetizing fields used varies little from the nrm mean direction. At first it appears this site has simply been remagnetized in recent times and its primary magnetization obliterated. However, detailed examination shows that after initial changes in sample directions of up to 15°, end-points are attained within the range 10–25 mT, followed by erratic changes increasing in size up to 30°. The precision parameter, k, decreases rapidly (Fig. 5) and then steadily increases to a pronounced peak at 25 mT. Thus, the samples appear to contain two distinct components—a secondary viscous component due to the earth's present field and, almost parallel to it, a harder component, which could be primary.

Discussion

The two whole-rock K–Ar ages indicate that the Savage Point sills are not associated with the Franklin diabases. However, the ages are compatible with a Mackenzie diabase age (1240 Ma). Mackenzie diabase is widely distributed throughout the

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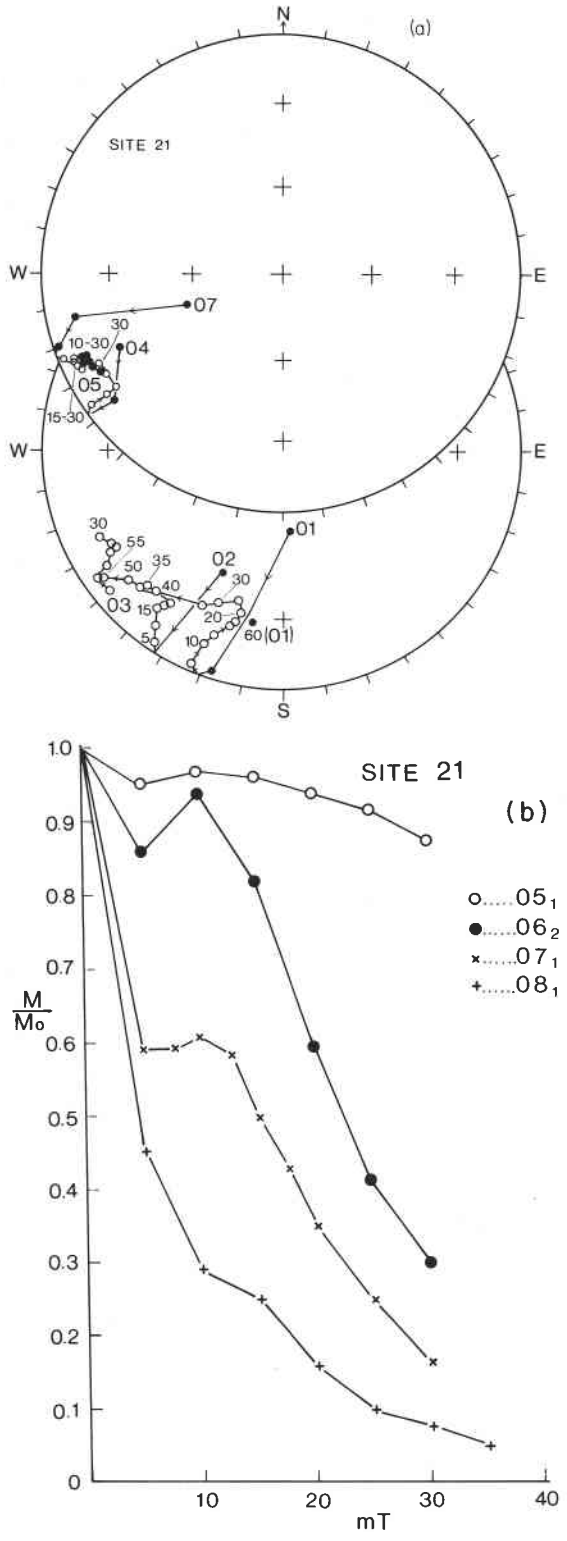


FIG. 4. (*a*) Changes in directions of magnetization on af demagnetization for specimens of site 21. Projection and convention as in Fig. 3. (*b*) Alternating magnetic field demagnetization curves for specimens of site 21.

Shield and although it has not previously been found in the Arctic Islands, Mackenzie dykes have been located (Fahrig and Jones 1969; Park 1974) in the Melville Peninsula, one of the northernmost parts of the mainland. The Savage Point mean pole before dip correction lies only 5.6° from that quoted by Irving and Hastie (1975) for all the Mackenzie diabase yet sampled (entry 1.453 of Irving and Hastie 1975). After dip correction the difference is 8.2°. Statistical comparisons between the two results show that they are indistinguishable at the 95% confidence level before the dip correction of the Savage Point results, but that on correction they become distinguishable. However, the latter case is marginal and a counter-clockwise rotation in the horizontal plane of less than 3° is required to make the two results indistinguishable once more. Rotations such as this of small blocks in the region of the Boothia Uplift are more than possible. It is suggested, therefore, that the Savage Point sills belong to the Mackenzie igneous episode.

The situation in the Aston Bay area is complex and it is possible that the paleomagnetic results embrace three different ages of igneous activity.

Site 21 intrudes the Aston Formation, is of normal polarity, and provides an uncorrected pole position nearly 22° from the Mackenzie diabase mean pole. If this diabase does not intrude the Hunting Formation then the comparison must be between the pole position corrected for dip and here the angular distance is nearly 27°. The correction has, however, been made using the dip and strike of the overlying Hunting Formation beds and may not reflect the total tectonic history of this diabase. A clockwise rotation of site 21 of 20° brings its corrected pole within 15° of the Mackenzie mean pole.

The poles from sites 19, 20, and 24 lie considerably to the west of the Mackenzie pole and are of reversed polarity. They lie closer to the mean Franklin diabase pole which includes both normal and reversed poles. Comparisons will be made with the mean pole of those Franklin diabases showing reverse polarity (entry 1.455 of Irving and Hastie 1975). The angular distance between the mean uncorrected pole of these sites and the Franklin pole is 14.0° and the two results are indistinguishable at the 95% confidence level. On correction for dip the three site poles are better grouped (the increase in K is not significant statistically) but their mean moves away from the Franklin pole (21.3°) and they become distinguishable. No amount of rotation in the horizontal plane will make the two results indistinguishable once more.

The Peuyuk kimberlite diatreme (intrusive into Upper Ordovician strata), further south on Somerset Island, gives a direction of magnetization of D =

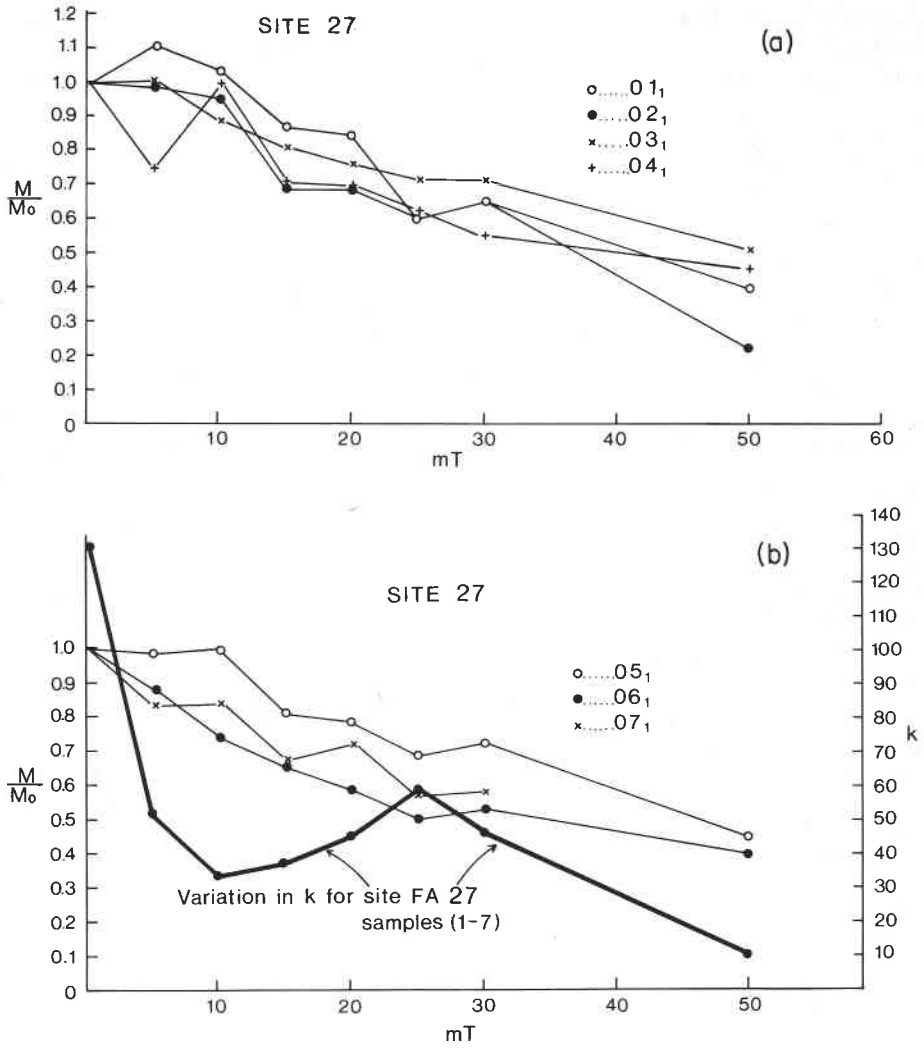


FIG. 5. (*a*) Alternating magnetic field demagnetization curves for specimens of site 27. (*b*) Additional af demagnetization curves and the variation of *k*, Fisher's precision parameter, with peak demagnetizing fields for site 27.

$36\frac{1}{2}°$, $I = 79°$ and this is thought by the present authors to be primary (this study will be reported separately). The corresponding virtual magnetic pole position is at 77.8°N, 3.5°W, approximately 15° from Tertiary paleomagnetic pole positions for North America (see, for example, McElhinny 1973). There are several other kimberlite occurrences on Somerset Island, presumably of the same age as the Peuyuk diatreme. There has thus been fairly widespread igneous activity on the island, certainly since the Upper Ordovician, possibly in the Tertiary. The harder magnetic component isolated at site 27 (where the lithology possibly indicates the present erosion surface is not a great deal below the original upper limit of the dyke) provides an *uncorrected* pole position at 73.9°N, 62.4°W, 14° from the Peuyuk pole and approximately 18° from Tertiary pole positions. It is not inconceivable, therefore, that the phase of igneous activity that gave rise to kimberlite emplacement also gave rise to the dyke in which site 27 is located. In this regard it is worth noting that the Cornwallis Fold Belt (part of the Boothia Uplift) has undergone block faulting during Cretaceous–Tertiary Eurekan Rifting (Kerr 1977).

An appraisal of geochronological, geological, and paleomagnetic results then suggests the following sequence of events in the Aston Bay area. Deposition and consolidation of the Aston Formation; intrusion of Mackenzie diabase identified via the sill-like mass of diabase (site 21); weathering and erosion which exposed this diabase (possibly accompanied by minor tilting); deposition and consolidation of the Hunting Formation followed by folding, faulting, and rotation; intrusion of Franklin dykes; and a lengthy period of weathering and ero-

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sion with the possible late intrusion of the large dyke (site 27) in Tertiary times.

In summary, the results from both Savage Point and Aston Bay suggest the Aston Formation is older than 1240 Ma, while those from the Aston Bay area can be interpreted as bracketing the Hunting Formation between 1240 and 675 Ma.

Acknowledgments

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